

DEPT OF ELECTRICAL & ELECTRONICS ENGINEERING

Program: T. Y. B. Tech ECE AI-ML Semester -6

Subject: Image Processing & Pattern Recognition (PC) (ECE3022B)

Experiment No.4

AIM: To perform edge detection using various masks in the spatial domain and to study edge linking, boundary representation

Introduction

Edges are significant local changes of intensity in a digital image. An edge can be defined as a set of connected pixels that forms a boundary between two disjoint regions. There are three types of edges:

- Horizontal edges
- Vertical edges
- Diagonal edges

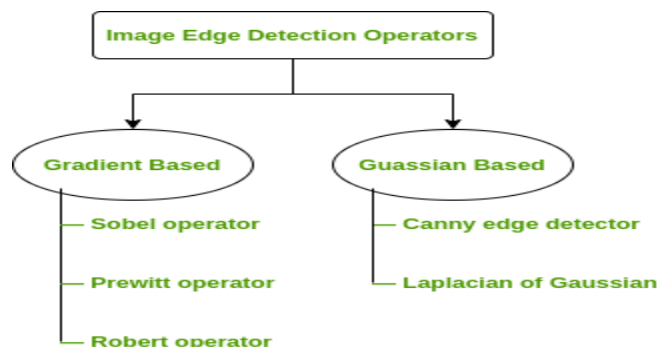
Edge Detection is a method of segmenting an image into regions of discontinuity. It is a widely used technique in digital image processing like

- pattern recognition
- image morphology
- feature extraction

Edge detection allows users to observe the features of an image for a significant change in the gray level. This texture indicates the end of one region in the image and the beginning of another. It reduces the amount of data in an image and preserves the structural properties of an image.

Edge Detection Operators are of two types:

- **Gradient-based** operator which computes first-order derivations in a digital image like, Sobel operator, Prewitt operator, Robert operator
- **Gaussian-based** operator which computes second-order derivations in a digital image like, Canny edge detector, Laplacian of Gaussian.



Sobel Operator: It is a discrete differentiation operator. It computes the gradient approximation of image intensity function for image edge detection. At the pixels of an image, the Sobel operator produces either the normal to a vector or the corresponding

gradient vector. It uses two 3×3 kernels or masks which are convolved with the input image to calculate the vertical and horizontal derivative approximations respectively –

Advantages:

Simple and time efficient computation

1. Very easy at searching for smooth edges

Limitations:

1. Diagonal direction points are not preserved always
2. Highly sensitive to noise
3. Not very accurate in edge detection
4. Detect with thick and rough edges does not give appropriate results

Prewitt Operator: This operator is almost similar to the sobel operator. It also detects vertical and horizontal edges of an image. It is one of the best ways to detect the orientation and magnitude of an image. It uses the kernels or masks –

Advantages:

1. Good performance on detecting vertical and horizontal edges
2. Best operator to detect the orientation of an image

Limitations:

1. The magnitude of coefficient is fixed and cannot be changed
2. Diagonal direction points are not preserved always

Robert Operator: This gradient-based operator computes the sum of squares of the differences between diagonally adjacent pixels in an image through discrete differentiation. Then the gradient approximation is made. It uses the following 2×2 kernels or masks –

Advantages:

1. Detection of edges and orientation are very easy
2. Diagonal direction points are preserved

Limitations:

1. Very sensitive to noise
2. Not very accurate in edge detection

Marr-Hildreth Operator or Laplacian of Gaussian (LoG): It is a gaussian-based operator which uses the Laplacian to take the second derivative of an image. This really works well when the transition of the grey level seems to be abrupt. It works on the zero-crossing method i.e when the second-order derivative crosses zero, then that particular location corresponds to a maximum level. It is called an edge location. Here the **Gaussian operator reduces the noise and the Laplacian operator detects the sharp edges.**

Advantages:

1. Easy to detect edges and their various orientations
2. There is fixed characteristics in all directions

Limitations:

1. Very sensitive to noise
2. The localization error may be severe at curved edges
3. It generates noisy responses that do not correspond to edges, so-called “false edges”

Canny Operator: It is a gaussian-based operator in detecting edges. This operator is not susceptible to noise. It extracts image features without affecting or altering the feature. Canny edge detector have advanced algorithm derived from the previous work of Laplacian of Gaussian operator. It is widely used an optimal edge detection technique. It detects edges based on three criteria:

1. Low error rate
2. Edge points must be accurately localized
3. There should be just one single edge response

Advantages:

1. It has good localization
2. It extract image features without altering the features
3. Less Sensitive to noise

Limitations:

1. There is false zero crossing
2. Complex computation and time consuming

Some Real-world Applications of Image Edge Detection:

- medical imaging, study of anatomical structure
- locate an object in satellite images
- automatic traffic controlling systems
- face recognition, and fingerprint recognition

The following program detects the edges of frames in a livestream video content. The code will only compile in linux environment. Make sure that openCV is installed in your system before you run the program.

Steps to download the requirements below:

- Run the following command on your terminal to install it from the Ubuntu or Debian repository.

```
sudo apt-get install libopencv-dev python-opencv
```

- OR In order to download OpenCV from the official site run the following command:

```
bash install-opencv.sh
```

- on your terminal.
- Type your sudo password and you will have installed OpenCV.

cv2.Sobel(): The function cv2.Sobel(frame,cv2.CV_64F,1,0,ksize=5)

- can be written as
cv2.Sobel(original_image,ddepth,xorder,yorder,kernelsize)
- where the first parameter is the original image, and the second parameter is the depth of the destination image. When ddepth=-1/CV_64F, the destination image will have the same depth as the source. The third parameter is the order of the derivative x. The

fourth parameter is the order of the derivative y. While calculating Sobelx we will set the x-order as 1 and y-order as 0 whereas while calculating Sobely, the case will be reversed. The last parameter is the size of the extended Sobel kernel; it must be 1, 3, 5, or 7.

- **cv2.Laplacian:** In the function `cv2.Laplacian(frame,cv2.CV_64F)`
- the first parameter is the original image and the second parameter is the depth of the destination image. When `depth=-1/CV_64F`, the destination image will have the same depth as the source.

Edge Detection Applications

- Reduce unnecessary information in an image while preserving the structure of image.
- Extract important features of image like curves, corners and lines.
- Recognizes objects, boundaries and segmentation.
- Plays a major role in computer vision and recognition

Algorithm

Part A

1. Read and display a grayscale image
2. Apply the Sobel operator in the X-direction and display the result
3. Apply the Sobel operator in the Y-direction and display the result
4. Determine and display images with X as well as Y directional edges.
5. Apply the Laplacian operator and display the result.

Part B

Study various algorithms for Edge Detection, Edge Linking and boundary representation. Implement any one of these algorithms.

Ex: Canny edge detection, edge linking using Hough Transform, Contours, Convex Hull, Harris Corners. Following are few online references:

contours

<https://learnopencv.com/contour-detection-using-opencv-python-c/>

Contours and convex hull

<https://medium.com/analytics-vidhya/contours-and-convex-hull-in-opencv-python-d7503f6651bc>

#Hough transform

<https://www.geeksforgeeks.org/line-detection-python-opencv-houghline-method/>

to find corners

<https://stackoverflow.com/questions/7263621/how-to-find-corners-on-a-image-using-opencv>

connecting edges

<https://stackoverflow.com/questions/65681384/connecting-ends-of-edges-opencv>

Edge and Harris corner

Questions

Q1. Apply the Sobel edge detector and Laplacian edge detector on the highlighted area and comment on the result.

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12	12	12	12	12
12	12	12	12	12
201	201	201	201	201
201	201	201	201	201

Note

- Write down the Expt Number, Title, PRN number, Name, and Date of performance in the first cell of the python code using markdown.
- Write down the “Result and Conclusion” in the cell next to the python code using markdown.
- Write down questions and answers in the last cell of the python code notebook using markdown.